CoachDash (Registry C)

Master Architecture & Operations Blueprint Version 1.0

1. The Coach Ecosystem Topology

CoachDash operates as the connective tissue between the physical real estate (GymDash/Clubdash) and the end consumer (Sodia). A coach is treated as a roaming micro-business. The system empowers them to build programming, monitor client biometrics, and manage scheduling without being confined to a single facility's proprietary software.

Ecosystem Vector	CoachDash Responsibility
To Facility (B2B)	Manages digital affiliations, floor availability windows, and verifies that clients have active gate access before allowing a session to be booked.
To Client (B2C)	Pushes custom macrocycles, workouts, and nutrition targets directly to the client's Sodia app. Tracks compliance and processes payments.
Internal (Ops)	Acts as a central CRM, video-analysis vault, financial ledger, and programming command line for the coach's independent business.

> **The Independence Principle:** Coaches own their Intellectual Property (custom drill libraries and macrocycles). If a coach leaves an affiliated facility, their client data, programs, and chat history move with them.

2. Harnessing the Sodia "Track" Module

CoachDash does not require the client to download a separate coaching app. Instead, CoachDash acts as a command-line interface that pushes and pulls data directly from the client's existing Sodia "Track" ecosystem (Food, Lift, Move).

Planned vs. Actual Logic

The Coach inputs targets into the `planned_track` table. The client executes and logs results in Sodia, which writes to the `actual_track` table. CoachDash automatically calculates the variance to flag non-compliance.

Track: Lift & Move

Coaches program sets, reps, RPE (Rate of Perceived Exertion), or distance blocks. These appear as a daily checklist on the client's Sodia app. Failed blocks instantly alert the coach.

Track: Food

Coaches set macro targets (Protein, Carbs, Fats) or binary habit checks ("Drink 3L Water"). Sodia's native food logger feeds this data back to the coach's compliance dashboard.

Automated Nudges

If the system detects a 3-day delta where `planned` exceeds `actual`, it auto-triggers a check-in message from the coach's profile to the client's Sodia inbox.

3. The Agnostic Programming Engine

Because CoachDash serves fitness trainers, tennis coaches, and running specialists alike, the programming builder cannot be hardcoded to just barbells.

- **Universal Drill Library:** A hybrid database. The system provides a master library of common movements, but coaches can upload their own custom videos and proprietary drills.
- **Cycle Builder:** A drag-and-drop calendar UI allowing coaches to construct Micro (weekly), Meso (monthly), and Macro (annual) cycles.
- **Agnostic Data Bank:** Just as Clubdash uses a JSON payload for varied sports scores, CoachDash stores performance metrics via JSON, allowing the frontend to render a 1RM chart for a powerlifter or a pace-curve for a marathoner seamlessly.

4. The Affiliation & Gate Matrix (Bulletproofing Access)

The greatest friction point in B2B2C coaching is the front-desk gate. CoachDash eliminates this by dynamically linking to facility access ledgers.

Scenario	System Logic & Resolution
The Expired Client	Coach attempts to book a session at "Gym A." The system queries GymDash's `entitlements` table. If the client's pass is expired, CoachDash blocks the booking and highlights the client in `GATE DENIED`, prompting a renewal push.
The Freelance Coach	If a coach books a session at a public park (no facility affiliation), the system intercepts the client's Sodia checkout with the Coach's personal independent liability waiver.
Floor Availability	Coach sets windows in CoachDash (e.g., "Tue 8-11 AM @ Gym A"). This actively updates GymDash's schedule, allowing the facility to forecast capacity and floor utilization.

> **Zero-Friction Entry:** When a valid session is booked, the client's Sodia QR code is automatically granted dual-authorization: it clears the facility gate (GymDash) and simultaneously checks them in for the coach's session ledger.

Financial Ledger & Payouts

CoachDash utilizes Stripe Connect to split payments. If a client buys a $100 "10-Session Pack," the money routes to the Coach. If the Coach's B2B Affiliation agreement requires a $10 "Floor Fee" per session, the system automatically fractionates that fee and routes it to the Facility (GymDash/Clubdash) upon session completion.

5. The User Flow: Day in the Life

The CoachDash UI is optimized for a tablet or mobile device, prioritizing immediate action over deep data diving during active floor hours.

- **06:00 AM - Triage & Adjust:** Coach opens the app. The "Needs Attention" queue flags three clients who missed their macro targets yesterday. The coach taps "Adjust Program" to lower their training volume for today's session.
- **08:00 AM - Floor Mode:** Coach switches to the *Live Roster* tab. It displays a vertical timeline of today's clients. As clients scan the facility gate, their status on the coach's tablet flips from "Expected" to `IN BUILDING`.
- **10:00 AM - Video Form-Check:** During a break, the coach opens the *Comms Hub*. A remote client uploaded a squat video. The coach scrubs the video, pauses it, uses a drawing tool to circle a posture error, and sends a timestamped voice note back to the client's Sodia inbox.
- **12:00 PM - Funnel Building:** Coach uses the *Offers* tab to publish a "4-Week Summer Boot Camp." The package is published directly to the Sodia consumer discovery feed, instantly accepting Apple Pay sign-ups.

> **The Client UX:** The client never leaves Sodia. They see their coach's face next to their daily tasks, receive push notifications for feedback, and use a single QR code for both gym entry and session check-in.

6. Business Analytics & CRM

A coach is a business. CoachDash provides macro-level analytics to ensure they are driving revenue and retaining their athlete base over time.

Compliance vs. Retention

Plots a client's "Track" completion percentage against their subscription length. Identifies the exact compliance drop-off point that precedes a client canceling their package.

Session Economy

Tracks active inventory: total outstanding prepaid sessions owed to clients versus sessions completed. Automatically triggers an upsell push to the client when they have 2 sessions remaining.

Venue ROI

Calculates net revenue generated across different affiliated facilities. Helps the coach decide if paying a high floor-fee at a premium gym yields a better return than training freelance at public parks.

The Data Bank Vault

Longitudinal tracking of client performance metrics. Visualizes progress over months and years, providing undeniable proof of the coach's efficacy to aid in contract renewals.